**ELLINGTON CYBER ACADEMY**

# Simulation 2 · Investigation Results

Normal administration and unusual executive access

**Learner**

Learner

**Case scope**

EHS-SIA-SIM-003 · 2026-10-02 · 18:00–18:30 UTC

**Time-window instructions**

Step 1 uses September 28–October 2, 2026, all five full UTC days. In Step 2, switch to October 2, 18:00–18:30 UTC and keep that focused window for Steps 2–4.

**Report generated (UTC)**

2026-11-11T02:30:03.699Z

**Knowledge-check result**

6/6 · 100.0% · Passed
Attempt: 1
Attempt submitted (UTC): 2026-11-11T02:10:54.987Z
Best score: 6/6

**Assessment scope**

Passing threshold: 80%. This is a browser-generated practice result, not an instructor-verified assessment. Written findings are not automatically graded.

# Investigation findings

### Which change reference supports Amara’s maintenance activity?

Based on the logs within the SIEM, we can see that EHS-IT-WS01 is performing approved maintenance on aokafor with a change reference of CHG-EHS-014. If we look in our documentation, we can see that EHS-IT-WS01 is within the asset inventory as our Windows IT service, and EHS-CHG-014 has approved changes from the Helpdesk maintenance for aokafor. This is also being initiated by an expected source IP address of 10.20.1.23.

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### Which user and source require further investigation?

Based on the evidence, cbennett requires further investigation. There are several things that stand out as anomalous, including:
- There are no approved changes listed for cbennett at this time.
- The source_ip is an unlisted public IP address (198.51.100.45) instead of the assigned private IP (10.20.1.51).
- The session_id is listed as s-bl-exec rather than s-bl-help like we see with aokafor.
- The action listed is application_access rather than approved_maintenance.
- The process_name and change_reference are both listed as null.

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### Which UTC event time supports that decision?

There are two events that pop up in this relevant 30-minute window:
- 2026/10/02 18:12:00.000 for the anomalous login from an unknown IP address
- 2026/10/02 18:14:00.000 for the anomalous application access from an unknown public IP

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**How many authentication records does each account have across the five-day baseline? Include your filters.**

user_name: _count
acarter: 24
amehta: 24
aokafor: 24
apatel: 23
cbennett: 25
dbrooks: 24
emartinez: 24
jblake: 24
mthompson: 24

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**Which event IDs distinguish authorized maintenance from the executive session?**

Event IDs BL-00 and BL-01 are associated with authorized maintenance while event IDs BL-02 and BL-03 are associated with executive sessions.

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**What should be escalated, and what does the evidence not prove?**

The activity in cbennett's account should be escalated for further review. The evidence doesn't fully prove malicious intent, as the logs only show an application was accessed but nothing more than that.

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**Record the baseline date/time range and any additional filters you used.**

This information was found between 2026/09/28 00:00:00.000 and 2026/10/02 23:59:59.999 and the query to find this information was
scenario_id="EHS-SIA-SIM-003" event_category="authentication"
| groupBy([user_name], function=count())

The earliest events occur at 2026/09/28 13:00:00.000 and the latest events occur at 2026/10/02 22:00:00.000

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**Supporting maintenance event IDs (separate with commas)**

BL-01

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**Executive-session evidence IDs or comparison notes**

BL-02 and BL-03 are both originating from an unlisted public IP (198.51.100.45)
BL-02 shows an executive session starting at 18:12:00.000 with a login with no process_name or change_reference.
BL-03 shows an executive session accessing an application at 18:14:00.000 with no process_name or change_reference.

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**Disposition for the maintenance activity**

Close as benign

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**Disposition for the executive session**

Escalate

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**Decision reasoning**

For aokafor, we can mark this as benign, as her account was approved for maintenance and all of the fields correctly point to the appropriate fields (the IP, the action, and the change_reference, as well as the event_ids)

For cbennett we're seeing executive logging, however their account is an EHS-EXEC-WS which doesn't automatically rule it out. However, we are seeing a login from an anomalous IP address and immediate access to ehs_finance, with a null process_name and change_reference. It's also worth noting that there were no failed login attempts within the timeframe.

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**Remaining unknowns**

For starters, we don't know if this information was malicious in nature, or if this was cbennett logging in from an unknown device, as the logs only show a login and application access. cbennett is associated with finance, which doesn't immediately make it stand out, but the IP, and null process_name and change_reference are red flags.

If this isn't cbennett, we don't know how they got their initial access, what they were truly able to access, if they were able to spread to any other devices, if there was any malware involved, if there was any data exfiltrated, and if there were any configuration changes.

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**Additional handoff notes**

To get a clean 5-day window of normal login behavior, I used the query:
scenario_id="EHS-SIA-SIM-003" event_category="authentication"
| groupBy([user_name], function=count())

This provided a clean picture of everyone's login, none of which stood out as abnormal at first glance.

To decide on aokafor, I used the query:
scenario_id="EHS-SIA-SIM-003" user_name="aokafor"
| sort(field=@timestamp, order=asc, limit=2000)
| table([@timestamp, event_id, user_name, source_ip, asset_name, session_id, action, outcome, process_name, change_reference, application])

This query gave a clear picture when compared to the asset inventory and change management logs, showing that this was all approved maintenance and green lit.

As for cbennett, the query used was:
scenario_id="EHS-SIA-SIM-003" user_name="cbennett"
| sort(field=@timestamp, order=asc, limit=2000)
| table([@timestamp, event_id, user_name, src_ip, asset_name, session_id, action, outcome, process_name, change_reference])

This provided the information, including the anomalous IP address and what was accessed. Within this 30 minute timeframe, there are no failed login attempts for cbennett, which could mean that it was simply shadow IT accessing his resources. More information is needed; regardless of the results of the anomalous IP, I suggest we escalate this as it's touching a financial system with an executive account.

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**Pinned evidence IDs**

BL-02, BL-03

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## Knowledge-check answers and feedback

**Which evidence best supports treating Amara’s maintenance as authorized?**

Selected: Her account and maintenance record match CHG-EHS-014.
Correct
Feedback: Amara’s approved maintenance is linked to CHG-EHS-014 in the public change register.
Correct response: Her account and maintenance record match CHG-EHS-014.

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**What needs further review in Charles’s focused case activity?**

Selected: His external-source session followed by finance application access.
Correct
Feedback: Charles’s external session and application access merit review; they do not alone prove compromise.
Correct response: His external-source session followed by finance application access.

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**To compare the five-day baseline with the focused October 2 activity, which local time scope should you choose first?**

Selected: Entire supplied dataset, covering September 28–October 2.
Correct
Feedback: The supplied SIM-003 dataset includes all five baseline days.
Correct response: Entire supplied dataset, covering September 28–October 2.

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**An event occurs outside a user’s normal schedule. What should an analyst do next?**

Selected: Compare source, session, and approval context before deciding.
Correct
Feedback: Baseline deviations need supporting context before a disposition is made.
Correct response: Compare source, session, and approval context before deciding.

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**Why group authentication records by account, source, and outcome?**

Selected: To compare login patterns across those field combinations.
Correct
Feedback: Grouping makes differences in source and outcome visible for each account.
Correct response: To compare login patterns across those field combinations.

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**Which disposition best reflects the two activities in this simulation?**

Selected: Explain approved maintenance separately from unusual executive access.
Correct
Feedback: Separate dispositions preserve the evidence that explains one activity and raises concern about the other.
Correct response: Explain approved maintenance separately from unusual executive access.

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## Search queries

**baseline**

```
scenario_id="EHS-SIA-SIM-003" event_category="authentication"
| groupBy([user_name], function=count())
```

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**maintenance**

```
scenario_id="EHS-SIA-SIM-003" user_name="aokafor"
| sort(field=@timestamp, order=asc, limit=2000)
```

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**executive**

```
scenario_id="EHS-SIA-SIM-003" user_name="cbennett"
| sort(field=@timestamp, order=asc, limit=2000)
```

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